MIZORAM PUBLIC SERVICE COMMISSION

COMPETITIVE EXAMINATION (COMBINED TECHNICAL) FOR RECRUITMENT TO THE POSTS OF AGRICULTURE & ALLIED SERVICES, GOVERNMENT OF MIZORAM.

(NOVEMBER-2025)

HORTICULTURE SCIENCE PAPER-III

Time Allowed : 3 hours FM : 200

SECTION - A *(Multiple Choice questions) (100 Marks)*

All questions carry equal mark of 2 each. Attempt all questions.

*This Section should be answered only on the **OMR Response Sheet** provided.*

Notes regarding Negative Marking :

(i) There will be penalty (negative marking) for every wrong answer given by a candidate in the MCQ type questions.

(ii) For each question for which a wrong answer has been given by the candidate, one-third of the marks assigned to that question will be deducted as penalty

(iii) If a candidate gives more than one answer to one question, it will be treated as a wrong answer even if any one of the given answers happens to be correct and there will be same penalty as above for that question.

(iv) If a question is left blank, i.e., no answer is given by the candidate, there will be no penalty for that question.

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1. A cybrid plant is formed when:

(a) Both nucleus and cytoplasm are fused

(b) Two haploid plants fuse naturally

(c) Only cytoplasm from one parent is combined with nucleus of another

(d) Mutation is induced by chemicals

2. A piece of potato is placed in a concentrated sugar solution for 30 minutes. After this, the potato becomes flaccid and shrinks. What is the reason?

(a) Water moved into the potato cells by osmosis.

(b) Water moved out of the potato cells by osmosis.

(c) Sugar moved into the potato cells by diffusion.

(d) Sugar moved out of the potato cells by diffusion

3. Break-even analysis in farm management is used to:

(a) Determine crop yield only

(b) Identify the level of production where total cost equals total revenue

(c) Calculate soil fertility

(d) Estimate labour requirement

4. *Bt* cotton is genetically engineered to:

(a) Resist herbicides

(b) Produce an insecticidal protein from *Bacillus thuringiensis*

(c) Tolerate drought conditions

(d) Increase seed oil content

5. Contract farming in horticulture primarily ensures:

(a) Fixed prices for inputs only

(b) Elimination of all production risks

(c) Free labour supply

(d) Guaranteed market and price for farm produce

6. Cryopreservation in plant biotechnology is usually performed at which of the following temperatures?

(a) 0–4°C (b) –20°C

(c) –80°C (d) –196°C

7. Deficiency of which nutrient causes interveinal chlorosis in young leaves?

(a) Nitrogen (b) Iron

(c) Magnesium (d) Manganese

8. In plant breeding, backcrossing is used to:

(a) Introduce a new trait from a different species

(b) Remove undesirable traits from a variety

(c) Increase the yield potential of a variety

(d) Develop prolific varieties

9. In plant breeding, the term heterosis refers to:

(a) Hybrid vigour leading to superior performance

(b) The process of self-pollination

(c) Genetic uniformity in a population

(d) The ability of a plant to adapt to various environmental conditions

10. In the context of NE India, Participatory Rural Appraisal (PRA) is particularly important because:

(a) Farmers prefer government-centric planning

(b) The region has diverse tribal knowledge and agro-ecological conditions

(c) Technology adoption is uniform across the region

(d) There is no need for local participation

11. In tissue culture media, the primary role of auxins is to:

(a) Induce shoot formation

(b) Promote root initiation and callus formation

(c) Prevent microbial contamination

(d) Maintain osmotic balance

12. KVKs are established at the:

(a) Village level (b) Block level

(c) District level (d) State level

13. One of the major challenges in implementing the T&V (Training and Visit) system in hill regions of NE India is:

(a) Lack of proven technologies

(b) Difficult terrain and dispersed villages

(c) Farmers unwillingness to learn

(d) Excessive financial resources

14. Protoplasts are plant cells:

(a) With intact cell wall (b) Without cell wall

(c) Only from roots (d) Only from meristematic tissue

15. Somaclonal variation in plant tissue culture refers to:

(a) Genetic variation arising from the sexual reproduction of cultured cells

(b) Genetic variation arising from the asexual reproduction of cultured cells

(c) Variation in the physical appearance of somatic cells

(d) Variation in the nutrient requirements of cultured cells

16. The bulk method in plant breeding is primarily used for:

(a) Developing pure lines

(b) Releasing new hybrid varieties

(c) Increasing the frequency of desirable alleles in a population

(d) Producing genetically identical clones

17. The core objectives of Viksit Krishi Sankalp Abhiyan (VKSA) 2025 is:

(a) Reducing cost of farming inputs to make high income for input providers

(b) Bridging the gap between agricultural research and field application

(c) Increasing the number of scientists interacting with farmers

(d) Focusing on Kharif season crops

18. The most common method for preservation of fruit juices, RTS and nectars, is:

(a) Sterilization (b) Pasteurization

(c) Freezing (d) Filtration

19. The father of modern genetics, Gregor Mendel, worked primarily on:

(a) Rice (b) Maize

(c) Pea (d) French Bean

20. The first Krishi Vigyan Kendra (KVK) in India was established in 1974 at:

(a) Ludhiana (b) Pondicherry

(c) Pune (d) Hyderabad

21. The first organized agricultural extension work in India before Independence began with which program?

(a) Intensive Agricultural District Programme (IADP)

(b) Community Development Programme (CDP)

(c) Grow More Food Campaign

(d) National Extension Service (NES)

22. The FSSAI 'Eat Right India' movement aims at:

(a) Improving food quality for export

(b) Raising awareness about healthy food choices

(c) Increasing the production of traditional foods

(d) Lowering food prices for consumers

23. The high osmotic pressure created by sugar in jam, jelly, and marmalade primarily helps to:

(a) Enhance the flavour and colour of the product

(b) Increase the pH of the fruit product

(c) Inhibit microbial growth and preserve the product

(d) Reduce the acidity of the fruit

24. The hormone primarily responsible for stomatal closure during water stress is:

(a) Gibberellin (b) Cytokinin

(c) Abscisic acid (d) Ethylene

25. The main advantage of a power tiller over manual tillage is:

(a) Requires no fuel

(b) Provides uniform ploughing and saves labour

(c) Can only be used in flat lands

(d) Does not require maintenance

26. The mechanism responsible for the ascent of sap in plants is:

(a) Root pressure (b) Capillarity

(c) Cohesion-tension theory (d) Active transport

27. The pedigree method of selection involves:

(a) Selecting plants based on their physical appearance

(b) Keeping detailed records of the ancestry of selected plants

(c) Allowing random mating among plants

(d) Using molecular markers to select plants

28. The principle of Learning by Doing in agricultural extension is most closely associated with:

(a) Demonstrations
(b) Farm journalism
(c) Rural sociology
(d) Farm mechanization

29. The process of removing field heat from harvested produce is known as:

(a) Pre-cooling
(b) Curing
(c) Ripening
(d) Fermentation

30. A common food preservative, potassium metabisulphite is commonly used as a stable source of:

(a) Hydrogen peroxide
(b) Salicylic acid
(c) Benzoic acid
(d) Sulphur dioxide

31. The spark ignition (SI) engine is commonly associated with which type of fuel?

(a) Diesel
(b) Petrol
(c) Kerosene
(d) Biogas

32. Which of the following is not responsible for colour change in fruits during ripening?

(a) Chlorophyll degradation
(b) Breakdown of carbohydrates
(c) Synthesis of carotenoids
(d) Synthesis of anthocyanin

33. Training of rural youth in NE India through Krishi Vigyan Kendras (KVKs) primarily aims to

(a) Provide agricultural skills and entrepreneurship
(b) Promote only traditional farming techniques
(c) Conduct purely laboratory-based research
(d) Import modern machinery

34. Which nutrient plays a key role in ATP formation and energy transfer in plants?

(a) Nitrogen
(b) Phosphorus
(c) Potassium
(d) Sulphur

35. Which of the following communication methods is most effective for women farmers in remote villages of Mizoram?

(a) Written reports and pamphlets
(b) Online webinars only
(c) Scientific journals
(d) Audio-visual aids and participatory demonstrations

36. Which of the following correctly defines the unit of power in SI system?

(a) Newton (N)
(b) Joule (J)
(c) Watt (W)
(d) Pascal (Pa)

37. Which of the following is a characteristic of a genetically pure line?

(a) High heterozygosity
(b) Uniformity in traits across generations
(c) High genetic diversity
(d) Ability to adapt to various environments

38. Which of the following is a characteristic of a successful entrepreneur in horticulture?

(a) Risk avoidance

(b) Proactive decision making and innovation

(c) Dependence on government subsidies only

(d) Resistance to change

39. Which of the following is a common method for controlling post-harvest diseases in fruits?

(a) Hot water treatment (b) Waxing

(c) Shrink wrapping (d) Precooling

40. In order to create modified atmosphere around fresh fruits and vegetables in modified atmosphere packaging, the type of packaging material used is:

(a) Freely permeable (b) Semi permeable

(c) Non permeable (d) Rigid plastic with air vents

41. Which of the following is a commonly used interculture implement?

(a) Disc harrow (b) Hoe

(c) Seed drill (d) Tractor-mounted plough

42. Which of the following is a method of pollination control in hybrid seed production?

(a) Protogyny (b) Protandry

(c) Male sterility (d) Cleistogamy

43. Which of the following is a non-climacteric fruit?

(a) Mango (b) Banana

(c) Pineapple (d) Papaya

44. Which of the following is a primary tillage implement?

(a) Cultivator (b) Harrow

(c) Plough (d) Seed drill

45. Which of the following is commonly used as a refrigerant in cold storage in India?

(a) R-134a (Tetrafluoroethane) (b) Ammonia (NH_3)

(c) Carbon dioxide (CO_2) (d) Water (H_2O)

46. Which of the following is NOT a method of gene transfer in plants?

(a) Agrobacterium-mediated transformation (b) Particle bombardment (biolistics)

(c) Electroporation (d) Grafting

47. Which of the following is the most commonly used basal medium for plant tissue culture?

(a) Murashige and Skoog (MS) medium (b) Gamborg B5 medium

(c) Whites medium (d) Hoagland solution

48. Which of the following is the primary purpose of anther culture in plant tissue culture?

(a) To induce somatic embryogenesis

(b) To produce haploid plants for breeding purposes

(c) To regenerate plants from leaf explants

(d) To study the effects of mutagens on plant cells

49. Which sugar is most commonly added to tissue culture media as a carbon source?

(a) Glucose (b) Fructose

(c) Sucrose (d) Lactose

50. Which of the following surface coating material for fruits & vegetables is obtained from shells of crustaceans like crabs, shrimps, etc?

(a) Chitosan (b) Carnauba wax

(c) Shellac (d) Alginates

Section - B *(Conventional Type) (100 Marks)*

*This Section should be answered only on the **Answer Sheet** provided.*
Marks for each question is indicated against it.
*Attempt **any 5 (five)** questions.*

1. What are the factors leading to post harvest loss of fruits and vegetables? Discuss the principles and methods of food preservation with special reference to fruits and vegetables. **(8+6+6=20)**

2. Discuss the genetic basis of qualitative and quantitative traits in plants. Explain the role of mutation and heterosis in crop improvement and illustrate with examples. **(4+16=20)**

3. Enlist the government schemes that support horticultural entrepreneurship in Mizoram. Explain how SWOT analysis can be applied to *(i) ginger (ii) birds eye chilli (iii) grapes (iv) passion fruit (v) pineapple* ventures in Mizoram. Suggest strategies for improving market access, export potential, and income generation. **(4+10+6=20)**

4. Evaluate the significance of Intellectual Property Rights (IPR) in safeguarding the agricultural knowledge and traditional crop varieties of ethnic communities in India. What challenges do these communities face? How can equitable benefit-sharing and conservation be ensured? **(8+4+8=20)**

5. Examine the limitations and achievements of plant breeding in North Eastern Hill (NEH) region of India. Discuss how modern molecular tools have enhanced the efficiency of breeding programs, with examples from major horticultural crops. **(8+12= 20)**

6. Explain the concept of Agri Export Zones (AEZs) and their importance in processed food exports. Evaluate the effectiveness of Hazard Analysis and Critical Control Point (HACCP) and Codex Alimentarius standards in improving the global competitiveness of Indian processed foods. **(10+10=20)**

7. How does water scarcity during the dry season impact the cultivation of vegetables in Mizoram? Critically evaluate integrated approaches to enhance horticulture crop resilience. **(6+14=20)**

8. What is the key difference between OFT (On-Farm Testing) and FLD (Front Line Demonstration) in terms of their objectives and application in agricultural technology refinement? How can extension activities improve farm productivity and resilience against climate and socio-economic challenges in North-East India? **(10+10=20)**

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